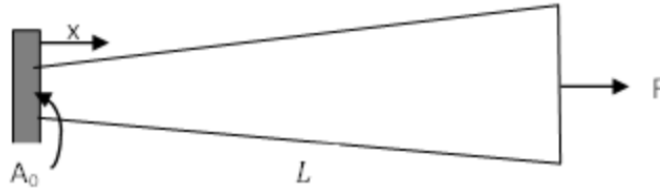


Instructions:

- Submit your assignment as a single PDF file through Canvas.
- Show all your steps and solution procedures.
- Make sure that your solution is well organised and that you are using appropriate headers for each question and sub-question.
- Scanned photos of your handwritten solution are acceptable as long as they are **legible**
- If a question requires plotting, please plot the expressions using any software/language (MATLAB, Excel, Mathematica, etc.) of your choice. However, **hand drawings** will not be accepted.
- You can use any software or calculator to perform any calculations. Using symbolic packages in MATLAB, Mathematica is acceptable as long as the solution steps are well laid out in your submission. In case you use those packages please attach the script in your submission file.
- While you can discuss the problems with classmates, all submitted work must be your own to conform to academic integrity guidelines.
- Late assignments will not be accepted.

Question 1 (80 points)

Consider a tapered bar of length L , Young's modulus E , and a variable cross sectional area $A(x) = A_0 \left(1 + \frac{x}{L}\right)$. The bar is fixed from one end and a concentrated force F is acting on the other end. The axial displacement u can be modelled using eq 1. The body force is assumed to be zero in this problem ($b(x) = 0$).



$$-\frac{d}{dx} \left(EA \frac{du}{dx} \right) = b(x) \quad (\text{eq 1})$$

- i. Derive the weighted integral statement of eq 1 (10 points)
- ii. Derive the weak form (20 points)

Given that: $E = 100 \text{ GPa}$, $A_0 = 0.005 \text{ m}^2$, $F = 100 \text{ N}$ and $L = 1 \text{ m}$

- iii. Estimate the solution using an $n = 1$ Ritz/Galerkin Approximation (20 points)

$$U_1(x) = \sum_{j=1}^1 c_j \theta_j(x) + c_o = c_1 x + c_o \quad \text{i.e. } \theta_1 = x$$

- iv. Estimate the solution using an $n = 2$ Ritz/Galerkin Approximation: (20 points)

$$U_2(x) = \sum_{j=1}^2 c_j \theta_j(x) + c_o = c_1 x + c_2 x^2 + c_o \quad \text{i.e. } \theta_1 = x \text{ and } \theta_2 = x^2$$

- v. Plot $u(x)$, $U_1(x)$ and $U_2(x)$ on the same figure. (10 points)

$$\text{Recall, from Assignment1 } u(x) = \frac{FL}{EA_0} \ln \left(1 + \frac{x}{L} \right)$$

Make sure that the axes are clearly labelled and include a legend.

NB: Please leave your answers for i & ii as symbolic expressions !

Question 2 (20 Points):

Please dedicate some time this weekend installing ABAQUS and select one of these choices

- a. I successfully installed Abaqus **(20 points)**
- b. I attempted installing Abaqus but faced difficulties and could not finish installing it **(20 Points)**
- c. I did not attempt installing ABAQUS **(0 Points)**

NB: You still get the full credit for attempting to install ABAQUS, please email me with the issues faced during the installation